

Inheritance and binary search

Inheritance

- extends: makes class a subtype of another
 - terminology: subclass and superclass
 - subclass has all attributes and methods of superclass
 - subclass can override methods it wants to change
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- access control
 - public: anyone can access it
 - private: can only access from within the class
 - protected: accessible from within class or a subclass

As a general rule, be as restrictive as possible

Which of the following lines has a syntax error?

```
class Person { private String name; }
```

```
class Student extends Person { //A
```

```
    private double credits;
```

```
    private double qualityPts;
```

```
    public double gpa() { return qualityPts / credits; }
```

```
    public String toString() {
```

```
        return name + " (GPA: " + gpa() + ")"; //B
```

```
    }
```

```
...
```

```
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```

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```
Person p = new Student(); //C
```

```
System.out.println(p.gpa()); //D
```

//E: Not exactly one of the above

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//E: Not exactly one of the above (B & D)

toString

- Method inherited from Object
 - Called when an object is printed
 - Comes with default implementation, but can be overridden by something more useful